

Annie Prasanna Manoharan

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Research Profile

M.S. Computer Engineering graduate with research experience using computer vision and deep learning to process and analyze annotated image and video data. Brings measurable model evaluation, reproducible preprocessing workflows, and medical-device software validation experience.

Education

University of Arkansas, Fayetteville, AR

M.S. Computer Engineering

Aug 2024 – May 2026

Coursework: Artificial Intelligence, Machine Learning, Image Processing, Data Mining

Sri Ramakrishna Engineering College, India

B.E. Electronics and Communication Engineering

Aug 2019 – May 2023

Technical Skills

Machine Learning & Vision: Python, PyTorch, deep learning, YOLO11n-Pose, OC-SORT, LSTM, Res-UNet, computer vision, image processing

Data & Research: SQL, Pandas, NumPy, data preprocessing, annotation pipelines, data validation, feature engineering, model evaluation

Software & Platforms: C++, C#, WPF/.NET, Flask, Git, Docker, LabVIEW, NI TestStand

ML / Data Tools: Scikit-learn, Apache Kafka, Apache Flink, ClickHouse, Grafana, data visualization, LangChain, FAISS

Projects

DualFit: Two-Stage Virtual Try-On

PyTorch, Res-UNet, flow-based warping, inpainting

- Developed a two-stage image pipeline combining flow-based garment warping with Res-UNet synthesis to improve alignment and visual realism.
- Built preprocessing and inpainting workflows, then trained and evaluated on VITON-HD: 24.9 PSNR, 0.913 SSIM, and 5.261 FID.

MiGa: Multi-Chicken Gait Assessment

YOLO11n-Pose, OC-SORT, LSTM, PyTorch

- Built a keypoint-estimation, multi-object-tracking, and many-to-one LSTM classification workflow for video analysis.
- Produced 4,569 annotated frames, 11,422 bounding boxes, and 55 expert-annotated sequences; achieved 91.83% keypoint mAP, 0.93 MOTA, and 0.91 IDF1.

Local LLM-Powered News Research Platform

Python, LangChain, FAISS, Ollama, Llama 3.2

- Built a local ETL and RAG workflow to ingest, clean, chunk, embed, and retrieve unstructured web content on demand.
- Used FAISS and nomic-embed-text for persistent vector storage and LangChain to orchestrate semantic queries over local Llama 3.2 models.

Professional Experience

Research Assistant, AICV Lab, University of Arkansas

Sep 2024 – Jan 2026

- Integrated YOLO11n-Pose, OC-SORT, and an LSTM network for multi-subject gait analysis from image and video data.
- Developed automated preprocessing and annotation workflows that transformed raw frames into structured model inputs and evaluation datasets.
- Trained and evaluated image-synthesis and tracking models using mAP, MOTA, IDF1, PSNR, SSIM, and FID metrics.

Engineering Intern / Project Engineer, Soliton Technologies

Jul 2022 – Jun 2024

- Designed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows for medical-device verification and validation.
- Implemented reusable object-oriented components and automated testing workflows to improve development and validation efficiency.
- Collaborated with cross-functional engineering teams to debug system-level issues, troubleshoot application behavior, and deliver production-ready software.

Teaching Assistant, University of Arkansas

Sep 2024 – May 2026

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging, and mentorship in C++ fundamentals.
- Supported Database Management Systems instruction for approximately 180 students, answering technical questions and assisting with assignments and assessments.